C L I P M I N D A I

Milestone 3 Report

Key Moments Detection and Analytics Dashboard

Video Summarization & Key Moments Detection Platform

Project	**ClipMind AI: AI-powered video transcription, summarization and key moments platform**
Report	Milestone 3 of 4 (Weeks 5 and 6)
Submitted by	Team 1
Domain	Artificial Intelligence
Document type	Project documentation (milestone report)

Topic segmentation, importance scoring, highlight clips, timestamp seeking and data-driven analytics.

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1. Milestone at a Glance

Milestone 3 turned the stored transcript into something a viewer can navigate. The system finds where topics change, scores how important each part is, selects the best moments, cuts a highlight clip for each one, and lets the user click a moment to jump to it in the video. An analytics dashboard summarises the processed data for creators and administrators.

Table 1: Summary of Milestone 3

Item	Details
Duration	Weeks 5 and 6
Focus	Key moments detection and analytics dashboard
Main deliverables	Transcript segmentation, Sentence Transformer embeddings, cosine similarity, topic segmentation, importance scoring, highlight selection with overlap removal, FFmpeg highlight clips, key-moments API, frontend Key Moments page with timestamp seeking, keyword extraction, analytics dashboards (creator and administrator), content insights
Technology used	Sentence Transformers (all-MiniLM-L6-v2), NumPy, FFmpeg, PostgreSQL, SQLAlchemy, Alembic, FastAPI, React
Tests	59 backend tests passed, 3 skipped, at this stage
Outcome	Key moments with topic, start and end time, text and score; clicking a moment seeks the video; analytics built from real database data

1.1 Objectives

- Treat the transcript as timestamped segments instead of one long paragraph.
- Use a pretrained embedding model and cosine similarity to find topic changes.
- Give each segment an explainable importance score between 0 and 1.
- Select useful highlights and avoid overlapping or tiny moments.
- Store video ID, topic, start time, end time, text and score for each key moment.
- Let the user click a timestamp and have the video start from that point.
- Build analytics from actual processed data, with explainable keyword extraction and no hard-coded values.

2. Timeline



Figure 1: Eight-week project timeline with Milestone 3 highlighted

Table 2: Week-by-week activities in Milestone 3

Week	Activities	Result
Week 5	Studied the Module 3 concepts (chunks, embeddings, similarity, topic segmentation, importance scoring). Built validation and chunking of Whisper segments. Added the embedding service and cosine similarity. Implemented topic-boundary detection and topic labels. Implemented importance scoring, selection, overlap removal and storage of key moments.	Key moments detected and saved
Week 6	Added FFmpeg highlight clips and saved the highlight path; added the migration and API. Built the Key Moments page with timestamp seeking. Built the analytics backend and dashboards. Integration review: fixed frontend and backend contract mismatches and video playback. Started the mentor's improvement round: quality, explainability and testing on different videos.	Key moments, clips and analytics available in the app

3. Concepts in Short

Table 3: Key concepts used in this milestone

Concept	Plain explanation	Where it is used
Segment / chunk	A small piece of the transcript with its own start and end time.	Each Whisper segment is analysed separately
Embedding	A list of numbers that represents the meaning of a text. Similar meanings give similar numbers.	all-MiniLM-L6-v2 model produces one vector per segment
Cosine similarity	A number that shows how close two embeddings are (1 means very similar, near 0 means unrelated).	Comparing each segment with the next one
Topic segmentation	Finding where the speaker moves from one subject to another.	A sudden similarity drop is a signal of a topic change, not proof
Importance score	A number from 0 to 1 that says how valuable a segment is.	Ranking segments for key moments
Keyword scoring	Words that are frequent in a few segments but not everywhere get a higher score (TF-IDF style).	Keyword coverage and topic labels
Overlap removal	If two selected moments cover mostly the same time, only the better one is kept.	Highlight selection
Highlight clip	A short video file cut from the original between a moment's start and end time.	FFmpeg clip per key moment
Aggregation	Counting and averaging stored data to show totals and trends.	Analytics dashboard

4. System Design

4.1 Key moment detection pipeline

Key moment detection works on the transcript that Whisper already produced. It does **not** run Whisper again and does **not** call an external AI service. The only model it loads is the local Sentence Transformer used for embeddings.

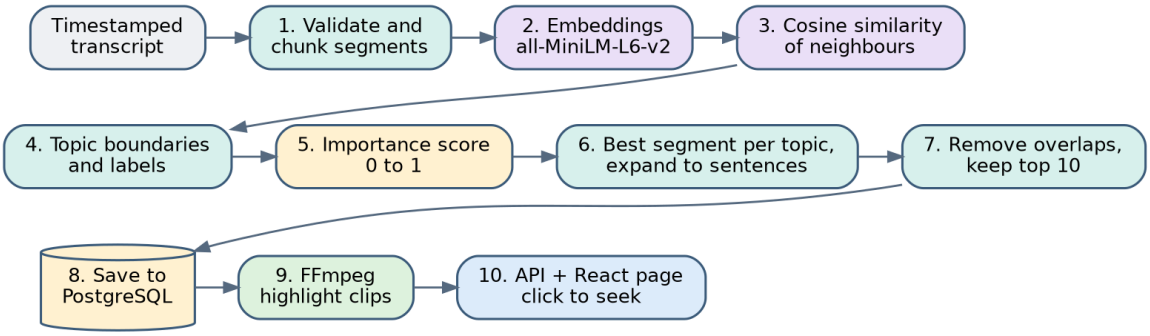


Figure 2: Key moment detection pipeline

4.2 Database design

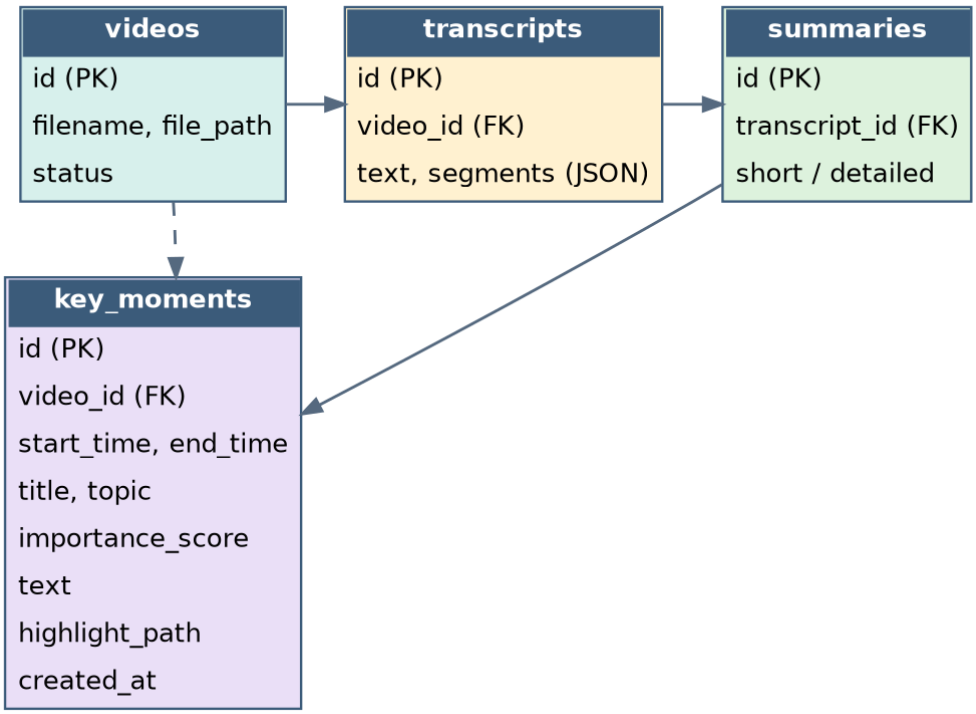


Figure 3: Entity relationships after Milestone 3

Table 4: The key_moments table

Field	Meaning
id	Primary key
video_id	The video this moment belongs to (foreign key)
start_time, end_time	Seconds from the start of the video, taken from Whisper timestamps
title	Short label built from the first sentence (at most 80 characters)
topic	Label of the topic region the moment belongs to
importance_score	Score from 0 to 1
text	Transcript text of the moment
highlight_path	Location of the highlight clip, exposed to the frontend as an API URL
created_at	When the moment was created (used by analytics)

5. Implementation: Key Moments

5.1 Step 1: Segmentation

Each Whisper segment is validated and kept with its original timestamps. Segments with invalid or reversed times, out-of-order starts, or no meaningful words are skipped. Timestamps are never invented or recalculated, which keeps every moment accurate against the real video.

5.2 Step 2: Embeddings and similarity

The text of every segment is converted to an embedding by the pretrained sentence-transformers/all-MiniLM-L6-v2 model (loaded once and cached; the model name can be changed with the EMBEDDING_MODEL setting). The cosine similarity between each segment and the next one is then calculated. If the embedding model cannot be used, the system falls back to one text-based topic region and never creates fake vectors.

5.3 Step 3: Topic segmentation

A low similarity between two neighbouring segments suggests a topic change. Because one dip can be noise, the rules require evidence before a boundary is accepted.

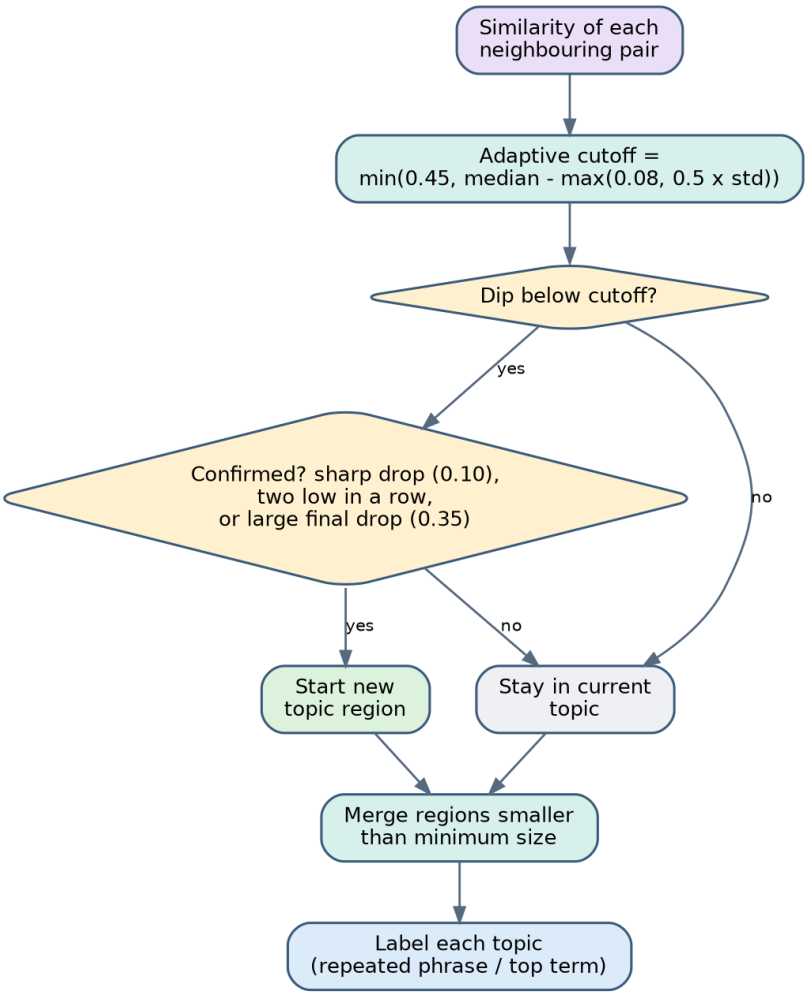


Figure 4: Topic boundary decision logic

Table 5: Topic segmentation parameters

Parameter	Value	Why
Base cutoff	0.45	Starting point for what counts as low similarity
Adaptive cutoff	min(0.45, median - max(0.08, 0.5 x standard deviation))	A generally low-similarity transcript does not turn every word shift into a new topic
Confirmation: sharp drop	At least 0.10 below both neighbours	A clear isolated change
Confirmation: sustained	Two low values in a row	A lasting change
Confirmation: final drop	At least 0.35 below the previous value	The last boundary has no following value to compare
Minimum region size	max(2, ceil(sqrt(number of segments) / 2)) segments; 1 for very short transcripts	Longer videos need more evidence for a new topic
Merge	Regions smaller than the minimum are merged into a neighbour	No tiny topics

Topic labels: each region is named from its own text. A repeated two-word phrase is preferred; otherwise the most salient single term is used. Filler words (for example "actually", "basically", "then") are removed, and the same label is not reused for two regions.

5.4 Step 4: Keyword extraction

Keywords are explained in plain terms. The transcript is split into segments, filler and stop words are removed, and each remaining word gets a score:

```
score(word) = count(word) x ( ln( (N + 1) / (df(word) + 1) ) + 1 )

N  = number of segments
df = number of segments that contain the word
```

A word that appears often, but in a limited number of segments, scores higher than a word spread evenly everywhere. The top 10 words are used as the video's keywords.

5.5 Step 5: Importance scoring

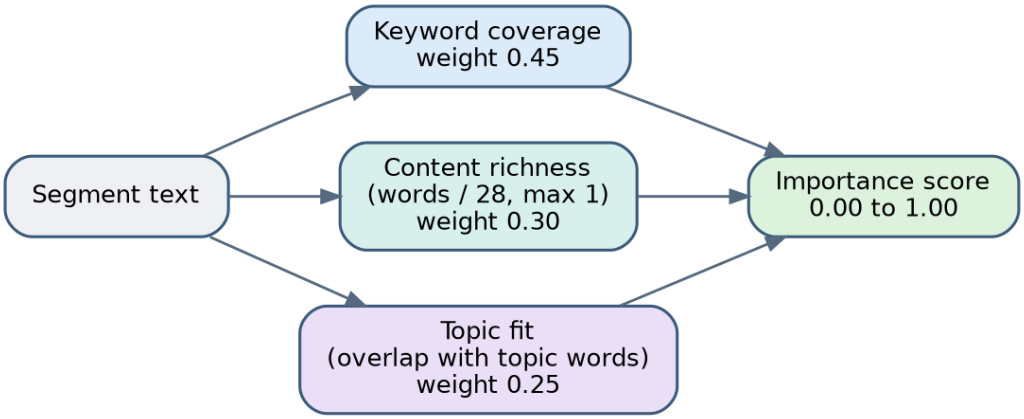


Figure 5: How the importance score is calculated

Table 6: Scoring components and reasoning

Component	Weight	Definition	Reasoning
Keyword coverage	0.45	Share of the video's top keywords found in the segment	Segments that carry the video's main terms are most informative
Content richness	0.30	Meaningful words in the segment divided by 28, maximum 1	Longer, substantive segments beat short chatter
Topic fit	0.25	Share of the segment's words that also appear in its topic region	Segments that represent their topic well are good highlights

5.6 Step 6: Selection and highlights

1. Within each topic region, the highest-scoring segment at or above the threshold (0.30) becomes a candidate.
2. If a candidate starts in the middle of a sentence, it is expanded to the nearest sentence boundaries, at most two segments each way and never across a topic region. Timestamps still come only from Whisper segments.
3. Overlaps are removed: if two moments overlap by 25% or more of the shorter one, only the higher-scoring moment is kept.
4. The top 10 moments by score are kept and sorted by time.
5. The moments are saved (replacing any older moments for the video).

Highlight clips: for each saved moment, FFmpeg cuts a clip between its start and end time (H.264 video, AAC audio). The clip location is stored as highlight_path. The API returns a URL such as /videos/5/highlights/11, not the server's file-system path.

6. Implementation: Frontend Key Moments Page

The page lists the key moments as cards and connects them to the video player.

- Each card shows the title, topic, start and end time, importance score and the transcript text.
- A **Play Moment** action seeks the player to the moment's start time.
- Loading, empty and error states are shown; moments appear only when processing is COMPLETED.
- Timestamps are returned in seconds as numbers, so the player can use them directly.

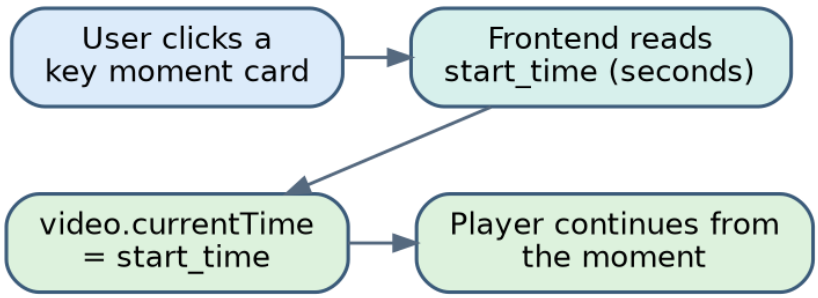


Figure 6: Clicking a key moment seeks the video

```
video.currentTime = keyMoment.start_time;
```

Video playback needed one fix. The media endpoint is protected by the JWT, but a plain HTML video tag cannot send an Authorization header. The frontend now fetches the media with the token and plays it from a temporary blob URL.

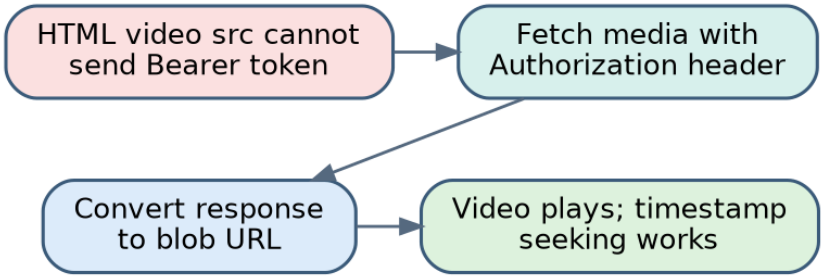


Figure 7: Authenticated video playback

7. Implementation: Analytics Dashboard

Analytics are computed from the real PostgreSQL data (videos, transcripts, summaries and key moments) with read-only queries. A creator sees only their own videos. An administrator sees the whole platform. Both can filter by date range.

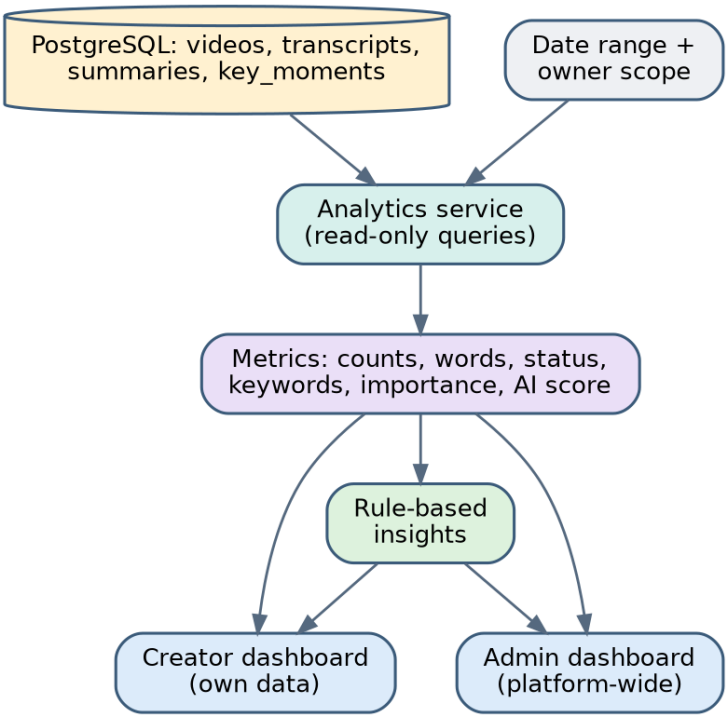


Figure 8: Analytics data flow

Table 7: What the dashboard reports

Section	Content
Overview and usage	Videos by status (completed, processing, failed, uploaded), uploads per day, users and roles (administrator view)
Transcript insights	Transcript count, total and average word count, characters
Summary reports	Summary count and completion, average summary length, compression

Section	Content
Key moment analytics	Total moments, videos with moments, total moment duration, importance distribution (Low up to 0.33, Medium up to 0.66, High above)
Keywords	Top 20 keywords with frequency, share of all keywords and number of videos
Topics	Most frequent content terms with related key-moment and summary counts
Recent videos	Per video: status, word counts, key-moment count, topics, importance and an AI score
AI insights	Up to five plain-language statements generated by rules from the data

Analytics keyword method

For analytics, words are lower-cased & counted across selected transcripts after removing stop words, filler words, numbers & words shorter than three letters. Result is ranked by frequency.

Video AI score

Table 8: How each video's AI score is calculated

Component	Weight	Scored from
Processing	25%	Video status (completed = 100)
Transcript	25%	Transcript status and word count
Summary	20%	Summary status and word count
Key moments	15%	Number of moments (20 points each, up to 100)
Content richness	15%	Distinct meaningful words and transcript length

The AI insights are produced by rules (for example, the leading topic, summaries that still need attention, processing still active). They are built from the actual numbers, not from a language model. Empty, loading, error and refresh states are handled on the page.

8. API Reference

Table 9: Endpoints added in Milestone 3

Method	Endpoint	Purpose
POST	/videos/{video_id}/key-moments/generate	Generate key moments from the existing transcript
GET	/videos/{video_id}/key-moments	Return key moments and processing status
GET	/videos/{video_id}/highlights/{moment_id}	Return the highlight clip of a moment (owner only)
GET	Creator and administrator analytics endpoints	Dashboard data with optional date range

```
GET /videos/5/key-moments
```

```
{
 "video_id": 5,
 "status": "completed",
 "key_moments": [{
  "id": 11, "start_time": 0.0, "end_time": 5.16,
  "title": "...", "topic": "Agents", "importance_score": 1,
  "text": "...", "highlight_path": "/videos/5/highlights/11"
 }]
}
```


9. Testing and Results

Table 10: Testing activities and outcomes

Area	What was checked	Outcome
Backend suite	All backend tests with the new module	59 passed, 3 skipped
Key moments and highlights	Detection, highlight generation, path saved in the database	Working
Ownership	Authentication ownership check preserved on the new endpoints	Passed
Migrations	New columns included in Alembic migrations	Included
Timestamp seeking	Click a moment and the video starts from that time	Passed
Different video types	Technical, educational, news and multi-topic videos	

10. Mentor Review and Improvements

Table 11: Improvement areas requested and actions

Mentor request	Action	Status
Make key moments meaningful with proper timestamps, topic, text and score	Sentence-boundary expansion, overlap removal, minimum thresholds, timestamps only from Whisper	Done; continued tuning
Improve topic segmentation and labels	Adaptive cutoff, evidence rules, minimum region size, filler-word filtering	Done; continued tuning
Explain how keywords are identified	TF-IDF style scoring documented (this report)	Done
Use actual processed data in analytics	Rebuilt on PostgreSQL queries, no hard-coded values	Done
Insights from real data	Rule-based insights computed from analytics numbers	Done
Usability of navigation between moments, topics, summaries and analytics	Key Moments page with seeking; role-based navigation	Ongoing
Test with different video types	Test matrix prepared (see Milestone 4 report)	In progress

11. Challenges and Resolutions

Table 12: Problems faced and how they were handled

Problem	Cause	Resolution	Status
Windows file path sent to the frontend	The database stored a server file path	API returns a URL (/videos/{id}/highlights/{moment_id}) as highlight_path	Resolved
Frontend and backend field mismatches	Key moment, summary and transcript fields differed between UI and models	Backend names kept; frontend maps fields instead of changing the API or schema	Resolved
Video would not play	A video tag cannot send the JWT to the protected media endpoint	Fetch with token and play from a blob URL	Resolved
Failed transcripts crashed pages	segments can be null when transcription fails	Frontend handles null safely	Resolved
Analytics did not match the backend	Different schema and authentication assumptions; router not registered	Analytics rewritten for the real models and registered correctly	Resolved
Topic noise and weak labels	Single similarity dips and filler words produced poor topics	Evidence rules, minimum region size, filler filtering	Improved
Moments starting mid-sentence	Whisper splits sentences across segments	Bounded expansion to sentence boundaries	Resolved
Analytics topics are frequent words	Analytics do not yet use the Module 3 topic labels	Planned: use real topic results for topic analytics	Open

12. Evaluation Against Milestone Criteria

Table 13: Milestone 3 evaluation checklist

Criterion	Status
Key moments detection and analytics dashboard implemented	Done
Highlight extraction workflows functional	Done
Content insights and reports generated	Done
Keyword extraction integrated	Done

13. Conclusion and Next Steps

Milestone 3 made videos navigable. Topics, key moments, highlight clips and analytics are all derived from one stored transcript, and every key moment can be explained: it comes from a measured topic region, a documented score and real timestamps.

Next (Milestone 4): add YouTube URL upload and the MCQ quiz, verify the full pipeline end to end, test across roles and video types, clean up migrations, prepare deployment and finalise documentation.

14. Appendix: Screenshots

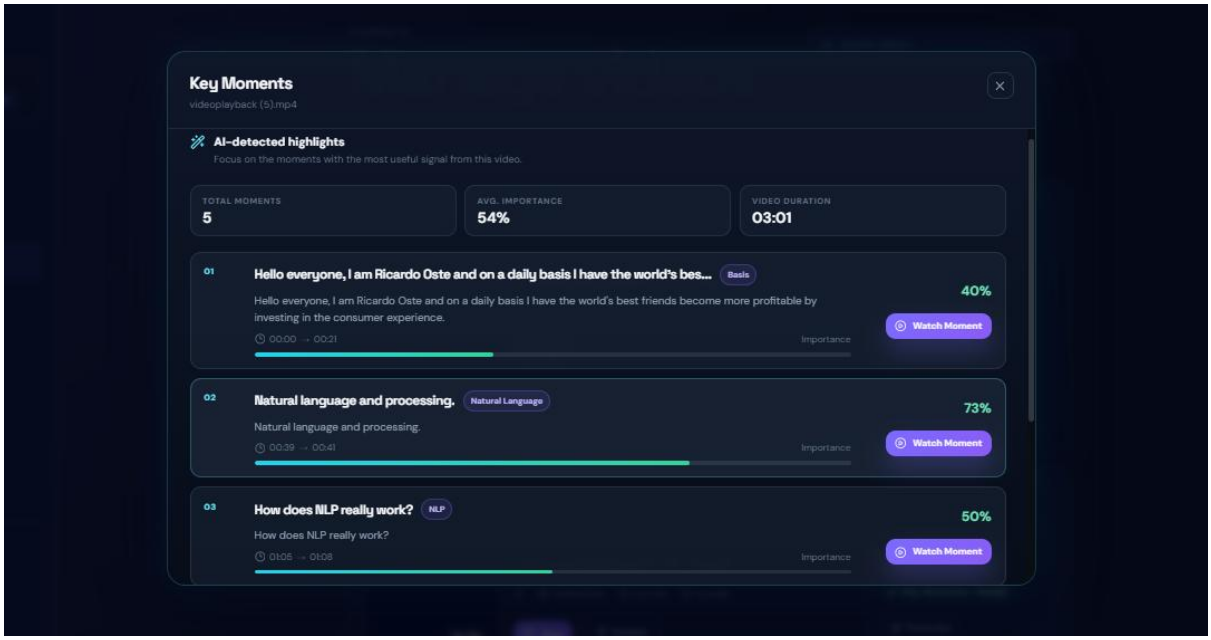


Figure 7: Key Moments page with topic, time, score and Play Moment

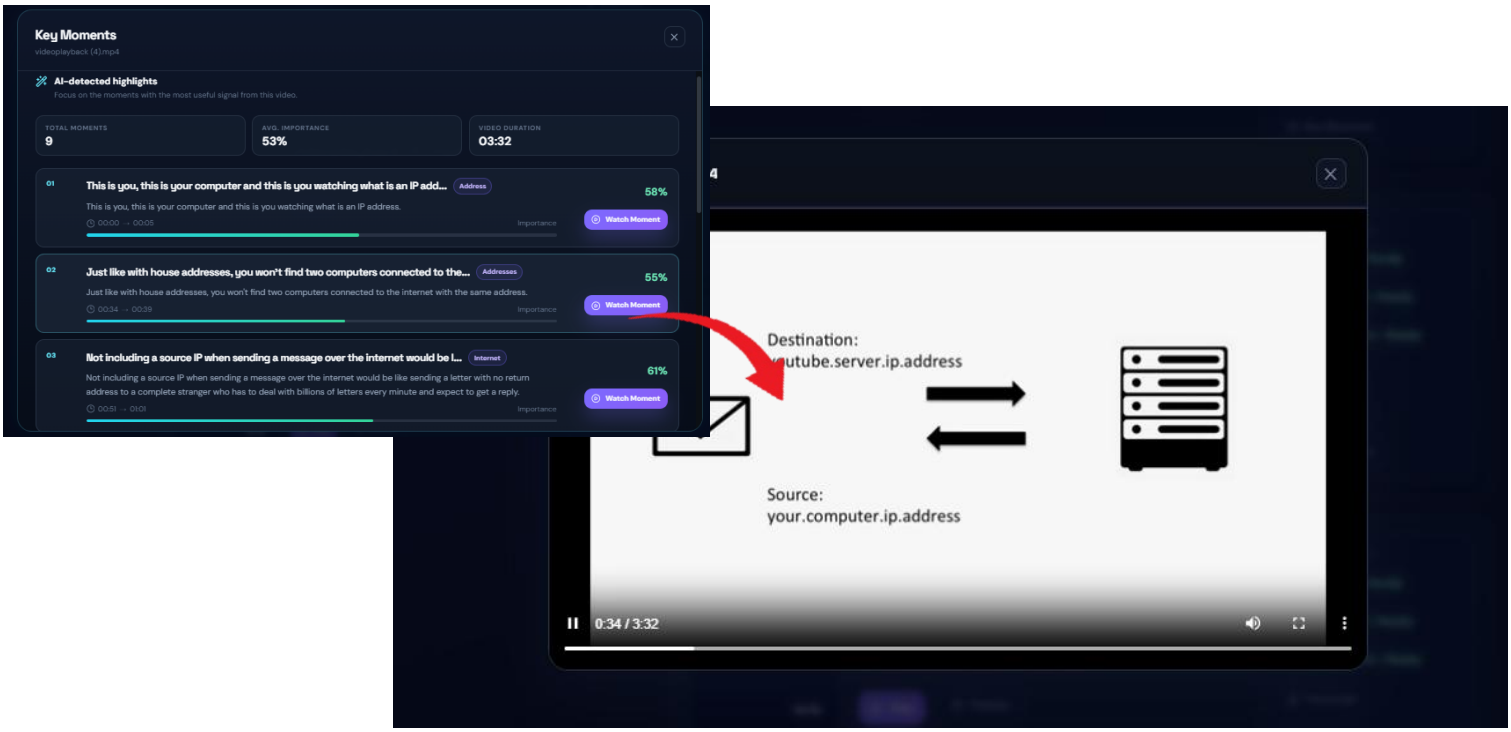
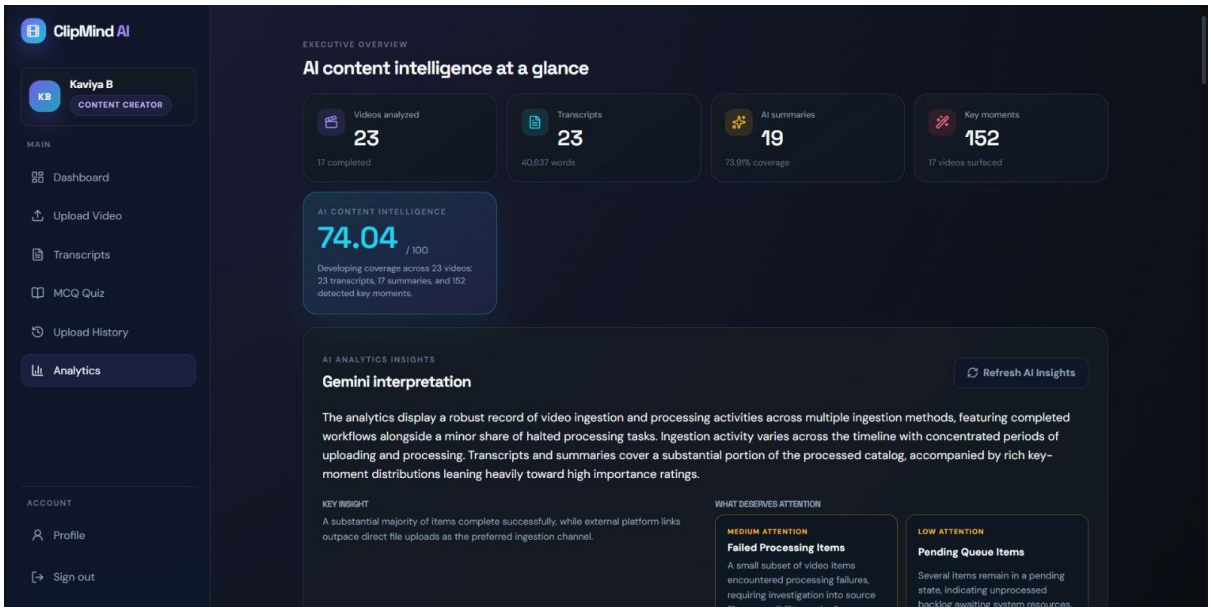


Figure 8: Video player after clicking a key moment


ClipMind AI
Kaviya B
CONTENT CREATOR
MAIN
Dashboard
Upload Video
Transcripts
MCQ Quiz
Upload History
Analytics
ACCOUNT
Profile
Sign out
EXECUTIVE OVERVIEW
AI content intelligence at a glance
Videos analyzed
23
17 completed
Transcripts
23
40,837 words
AI summaries
19
73.91% coverage
Key moments
152
17 videos surfaced
AI CONTENT INTELLIGENCE
74.04
/ 100
Developing coverage across 23 videos: 23 transcripts, 17 summaries, and 152 detected key moments.
AI ANALYTICS INSIGHTS
Gemini interpretation
Refresh AI Insights
The analytics display a robust record of video ingestion and processing activities across multiple ingestion methods, featuring completed workflows alongside a minor share of halted processing tasks. Ingestion activity varies across the timeline with concentrated periods of uploading and processing. Transcripts and summaries cover a substantial portion of the processed catalog, accompanied by rich key-moment distributions leaning heavily toward high importance ratings.
KEY INSIGHT
A substantial majority of items complete successfully, while external platform links outpace direct file uploads as the preferred ingestion channel.
WHAT DESERVES ATTENTION
MEDIUM ATTENTION
Failed Processing Items
A small subset of video items encountered processing failures, requiring investigation into source
LOW ATTENTION
Pending Queue Items
Several items remain in a pending state, indicating unprocessed


UPLOAD SOURCE BREAKDOWN
Video sources
File Uploads
YouTube URL Uploads
SUMMARY REPORTS
Coverage and status
58.33%
Summary coverage
12
Total Summaries
7
Completed Summaries
5
Not Completed
12
Videos with Summaries
CONTENT INSIGHTS
Topics and transcript richness
network
Most Discussed Topic
10
Topics Surfaced (up to 10)
3,252
Transcript Words
666 words
Longest Transcript
55
Key Moments
12
Videos with Key Moments
CONTENT DISTRIBUTION BY SURFACED TOPIC
network
0.95%
address
0.95%
KEYWORD INTELLIGENCE
Meaningful terms by frequency
network 17
address 17
hadoop 15
data 15
one 14
computer 14
really 13


USAGE STATISTICS
Activity for the selected period
12
Total Uploads
8
Processed Videos
12
Summaries Generated
55
Key Moments Generated
0
Users
UPLOAD ACTIVITY
PROCESSING ACTIVITY
10-14
10-15
10-14

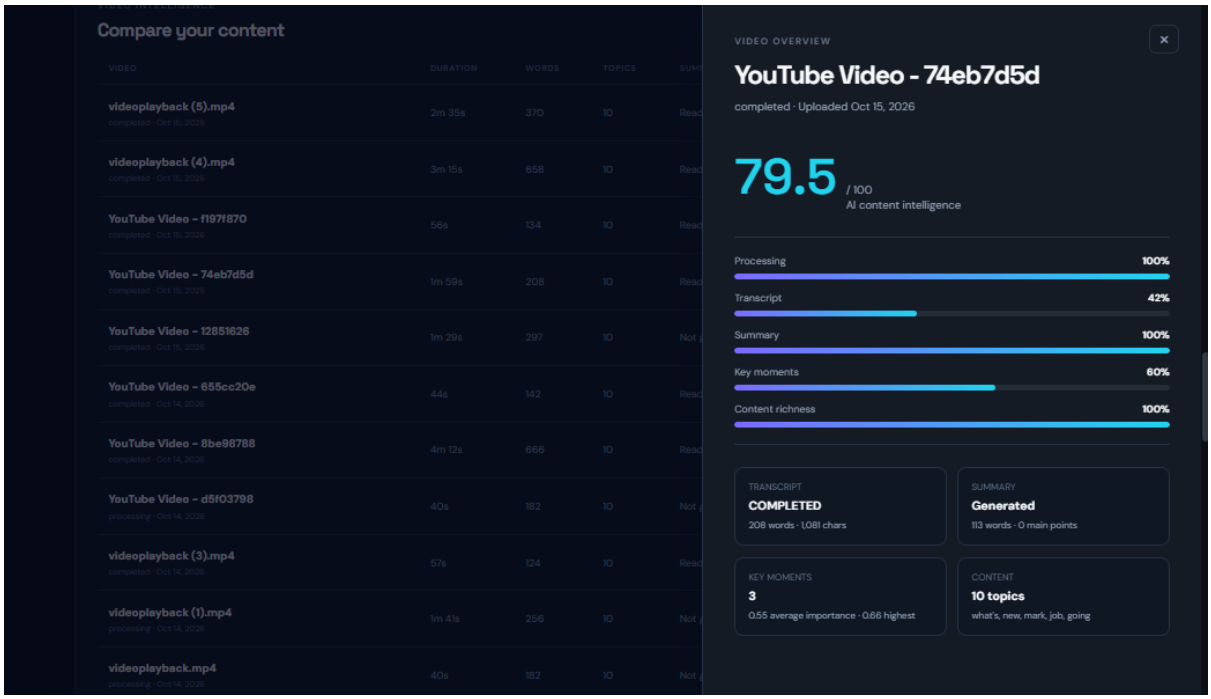

Compare your content
VIDEO
DURATION
WORDS
TOPICS
videoplayback (5).mp4
2m 35s
370
10
videoplayback (4).mp4
3m 15s
658
10
YouTube Video – f197f870
56s
134
10
YouTube Video – 74eb7d5d
1m 59s
208
10
YouTube Video – 12851626
1m 29s
297
10
YouTube Video – 655cc20e
44s
142
10
YouTube Video – 8be98788
4m 12s
686
10
YouTube Video – d5f03798
40s
182
10
videoplayback (3).mp4
57s
124
10
videoplayback (1).mp4
1m 41s
256
10
videoplayback.mp4
40s
182
10
VIDEO OVERVIEW
YouTube Video - 74eb7d5d
completed · Uploaded Oct 15, 2026
79.5
/ 100
AI content intelligence
Processing
100%
Transcript
42%
Summary
100%
Key moments
60%
Content richness
100%
TRANSCRIPT
COMPLETED
208 words · 1,081 chars
SUMMARY
Generated
113 words · 0 main points
KEY MOMENTS
3
0.55 average importance · 0.66 highest
CONTENT
10 topics
what's, new, mark, job, going



Figure 9: Administrator analytics dashboard

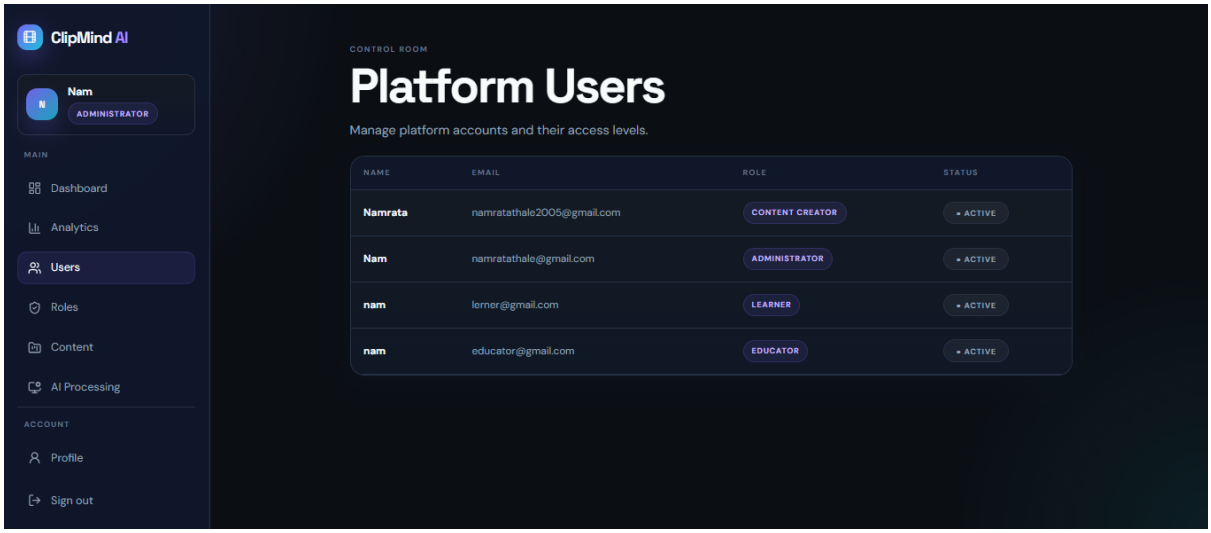


Figure 10: Administrator Registered users page



Figure 11: Administrator Roles & Permission details page

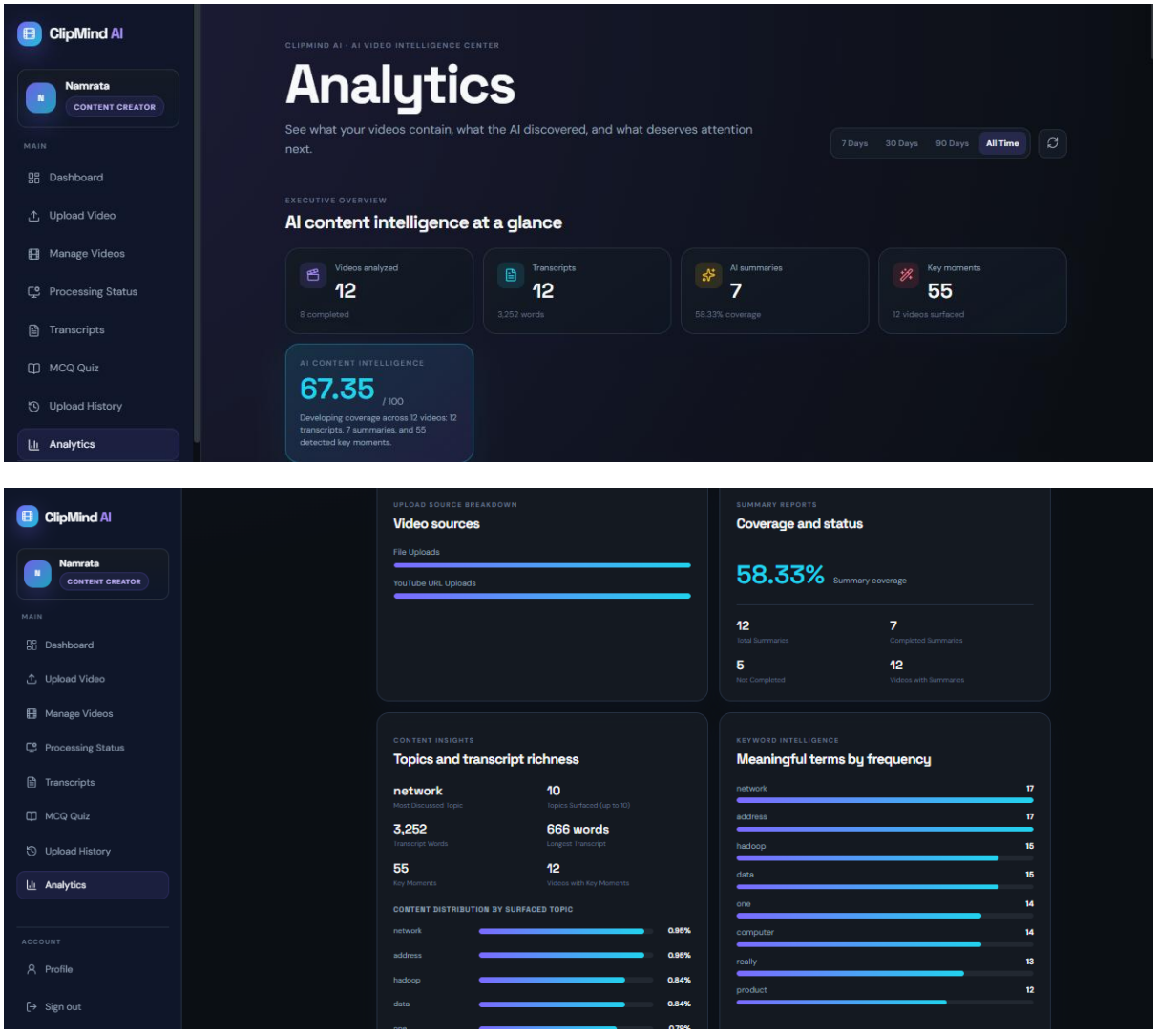


Figure 12: Creator analytics dashboard